Francisco Pinto Berkemeier

Curriculum Vitae

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Cambridge, United Kingdom
Nationality: Portuguese
DPOB: Macau, July 22, 1993

Research & Education

Sep 2022 – Present **Postdoctoral Researcher**, University of Cambridge, Departments of Pathology & Genetics.
Postdoctoral position in the Boemo Group, University of Cambridge. Supervisor: Prof. Michael A. Boemo.
Research topics: mathematical modelling and physics-informed machine learning to study genome-wide DNA replication dynamics, with a focus on replication stress, cancer-associated perturbations, and therapeutic response.

Sep 2018 – Nov 2022 **Doctor of Philosophy**, University College London, Department of Mathematics.
PhD in Mathematics. Supervisor: Prof. Karen M. Page. Second supervisor: Prof. Nadia Sidorova.
Research topics: mathematical modelling of the Notch-Delta signalling pathway and evolutionary dynamics.
Thesis title: *Cell-cell interactions in epithelial patterning: Notch-Delta signalling and evolutionary dynamics*.
Award date: 28 November 2022.

Aug 2016 – Jun 2018 **Master of Science**, King Abdullah University of Science and Technology, KSA.
MSc in Applied Mathematics and Computational Sciences. Supervisor: Prof. Diogo A. Gomes.
Thesis title: *A priori regularity of parabolic partial differential equations*.
Defense date: 10 April 2018.

Sep 2012 – Jun 2016 **Bachelor of Science**, Instituto Superior Técnico, University of Lisbon.
BSc in Applied Mathematics and Computation (LMAC).

Teaching Experience

Sep 2025 – Present **Guest Lecturer and Examiner**, Departments of Pathology & Genetics, University of Cambridge.
Lecturer for Modelling with PDEs in Computational Biology, as part of the NST Part III Systems Biology course, introducing reaction–diffusion models in biology. The lectures cover the origin of the equations, analytical and numerical approaches to their solution, and key applications in population dynamics, pattern formation, and tissue modelling. Contributed to exam preparation for the course. In parallel, served as examiner for first-year PhD progression vivas, assessing research plans, methodology, and scientific development.

Mar 2026 – Present **College Research Associate**, Emmanuel College, University of Cambridge.
Early-career College research affiliation, contributing to the academic and interdisciplinary life of the College through participation in scholarly events, teaching, and engagement with the wider collegiate community.

Oct 2023 – Oct 2025 **Postdoctoral Research Associate and College Supervisor**, Queens' College, University of Cambridge.
Rokos PDRA at Queens' College, with responsibility for supervisions in Part 1A of the Cambridge Computer Science Tripos and Natural Sciences Tripos, including the following modules and topics: Mathematical Methods I: Vector Calculus, Differential and Integral Calculus, Probability and Statistics; Mathematical Methods II: Ordinary and Partial Differential Equations, Multivariable Calculus, Applications to Physical Systems.

Aug 2020 – Jan 2021 **Lecturer**, Department of Mathematics, University College London.
Lecturer for MATH0049 - Mathematics for Engineers 1, covering topics such as differentiation, integration, ordinary differential equations, vectors, probability, statistics, and numerical methods. Responsibilities included delivering lectures, preparing exams and homework assignments, and marking.

Sep 2018 – Sep 2022 **Teaching Assistant**, Department of Mathematics, University College London.
Provided tutorial support, exam/homework marking, and office hours for undergraduate modules: Calculus and Linear Algebra (MATH0045); Analysis 1 & 2 (MATH0003 & MATH0004); Mathematical Methods 1, 2, & 3 (MATH0010, MATH0011& MATH0016); Newtonian Mechanics (MATH0009).

Sep 2017 – Jun 2018 **Teaching Assistant**, Applied Mathematics, King Abdullah University of Science and Technology.
Delivered tutorials, graded assignments, and supported Master's-level courses, including: Vector Calculus and Differential Equations (AMCS 131); Linear Algebra (AMCS 151); Applied Numerical Methods (AMCS 152); Engineering Mathematics (AMCS 101).

Sep 2015 – May 2016 **University Tutoring**, Pure and Applied Mathematics, Lisbon, Portugal.
Tutored university students through tutoring centres and the IST mathematics student association (NMATH), focusing on calculus, linear algebra, and mathematical analysis.

Research Supervision

Oct 2026 – Dec 2026 **PhD Research Supervision**, Department of Pathology, University of Cambridge.
Frederico Agua, PhD student, UCIBIO, NOVA School of Science and Technology. Supervision of a visiting PhD researcher as part of the PARSUK–FCT Bilateral Research Fund project *Decoding DNA replication across the cancer epigenome*. The project develops multimodal machine-learning approaches integrating genomic sequence with epigenomic data to investigate the regulation of DNA replication initiation across cellular contexts.

Jul 2026 – Aug 2026 **Undergraduate Research Supervisor**, Department of Pathology, University of Cambridge.
Ben Waldron, Natural Sciences Tripos undergraduate, University of Cambridge. Supervised a third-year summer research project on the development of Fourier Neural Operators for modelling DNA replication kinetics. Provided guidance on mathematical model formulation, numerical simulation, training-data generation, neural-operator implementation, and the biological interpretation and communication of results.

Jun 2023 – Jun 2024 **Master's Supervisor**, Department of Pathology, University of Cambridge.
Rosie Berners-Lee, Master's student, University of St Andrews, undertaking a year-long research placement at the Department of Pathology, University of Cambridge. Supervised a research project in DNA replication modelling. The work resulted in a dissertation awarded the highest distinction (20/20) and led to a peer-reviewed publication [6].

Research Funding Awards, Prizes & Distinctions

2026 **PARSUK-FCT Bilateral Research Fund (BRF)**, Competitive research grant awarded for *Decoding DNA replication across the cancer epigenome*, a UK–Portugal collaborative project between UCIBIO, NOVA School of Science and Technology, and the University of Cambridge.

2026 **Best Talk Award, runner-up**, Pathology Annual Symposium, University of Cambridge. Awarded for the plenary talk *A genome language model for mapping DNA replication origins*.

2025 **Nature Communications Editors' Highlights**, The article *DNA replication timing reveals genome-wide features of transcription and fragility* [5] was selected for Nature Communications Editors' Highlights in Genetics, Genomics and Epigenetics, recognising the 50 most outstanding recent publications in the field.

2025 **Best Poster Prize**, Research in Genetics Day 2025, Department of Genetics, University of Cambridge. Awarded for the poster *A computational framework for mapping DNA replication kinetics*.

Publications

[1] Huang, X., Berkemeier, F., Berners-Lee, R., Sherborne, K., Chapman, S.E., Merrick, C.J., and Boemo, M.A. (2026). Nanopore sequencing reveals the DNA replication timing programme of the malaria parasite *Plasmodium falciparum*. *bioRxiv*. doi:10.64898/2026.10.12.758579.

[2] Alkhaled, A., Berkemeier, F., Boemo, M.A., and Nik, K. (2026). Completion of DNA replication is constrained by the spatiotemporal organisation of origin firing. *arXiv (under review)*. doi:10.48550/arXiv.2609.07924.

[3] Pfuderer, P.L., Berkemeier, F., Nassar, J., Crisp, A., Jaworski, J.J., Moore, J., Sale, J.E., and Boemo, M.A. (2026). A genome language model for mapping replication origins. *bioRxiv (under review)*. doi.org/10.64898/2026.01.29.702604.

[4] Paci, G.*, Berkemeier, F.*, Baum, B., Page, K.M., Mao, Y. (2026). 3D epithelial cell topology tunes signalling range to promote precise patterning. *PNAS*. doi:10.1073/pnas.2522727123
*Joint first authorship.

[5] Berkemeier, F., Cook, P.R., and Boemo, M.A. (2025). DNA replication timing reveals genome-wide features of transcription and fragility. *Nature Communications*. doi:10.1038/s41467-025-59991-w.
★ Selected for Nature Communications Editors' Highlights in Genetics, Genomics and Epigenetics.

[6] Berners-Lee, R., Gilmore, E., Berkemeier, F.†, and Boemo, M.A. (2025). Regulation of replication timing in *Saccharomyces cerevisiae*. *PLOS Computational Biology*. doi:10.1371/journal.pcbi.1013066.

[7] Berkemeier, F., and Page, K.M. (2023). Coupling dynamics of 2D Notch-Delta signalling. *Mathematical Biosciences*. doi:10.1016/j.mbs.2023.109012.

[8] Berkemeier, F., and Page, K.M. (2023). Unifying evolutionary dynamics: a set theory exploration of symmetry and interaction. *bioRxiv*. doi:10.1101/2023.09.27.559729.

[9] Berkemeier, F. Cell-cell interactions in epithelial patterning: Notch-Delta signalling and evolutionary dynamics. PhD Thesis, University College London (2022). discovery.ucl.ac.uk/id/eprint/10158781.

[10] Berkemeier, F., and Gomes, D.A. (2019). A Priori Regularity of Parabolic Partial Differential Equations. Chapter in *New Trends in Analysis and Geometry*, *Cambridge Scholars Publishing*. ISBN:978-1-5275-4563-2.

Legend †*Corresponding author.*

Software

[S1] Berkemeier, F. (2026). *RepliCanvas: A browser-based studio for creating publication-ready DNA replication diagrams and animations*. fberkemeier.github.io/RepliCanvas.

[S2] Pfuderer, P. L., Berkemeier, F., and Boemo, M. A. (2026). *ORILINX: ORIgin of replication Language-model Inference via Nucleotide conteXt*. github.com/Pfuderer/ORILINX.

[S3] Berkemeier, F. (2026). *FNO-replication: A Physics-Informed Fourier Neural Operator (FNO) for learning and predicting DNA replication kinetics*. github.com/fberkemeier/FNO-replication.

[S4] Berkemeier, F. (2025). *Multi-layer Signalling Model (MSM): Simulations of Notch-Delta signalling in 3D epithelial tissues*. github.com/fberkemeier/MultiLayer-NotchDelta.

[S5] Berkemeier, F. (2025). *A toolkit for analyzing DNA replication timing, origin firing rates, and genomic stability across cell lines and genomic regions*. github.com/fberkemeier/DNA_replication_model.

[S6] Berkemeier, F. (2022). *Interactive Epithelium: a Mathematica tool for Notch-Delta 2D epithelial signalling simulations* (Version 1.0.0). github.com/fberkemeier/Notch-Delta-Coupling.

Talks, Conferences & Workshops

Jul 2026 **ECMTB 2026**, European Conference on Mathematical and Theoretical Biology, University of Graz. Contributed talk in the Mathematical Oncology session, entitled *A mathematical framework for replication stress in cancer*.

Jul 2026 **GenomePT 2026**, Facuçty of Sienices, University of Lisbon, Invited talk entitled *A genome language model for mapping DNA replication origins*.

Jun 2026 **Champalimaud Foundation**, Invited Seminar, hosted by Dr. Rita Fior, entitled *Towards a multimodal framework for mapping DNA replication kinetics*.

May 2026 **Cambridge AI Club for Biomedicine**, University of Cambridge, Jeffrey Cheah Biomedical Centre. Talk entitled *A genome language model for mapping DNA replication origins*.

Apr 2026 **Seminário de Matemática Aplicada e Análise Numérica**, Department of Mathematics, Instituto Superior Técnico, Lisbon. Research seminar entitled *Inverse problems and ML in DNA replication*.

Mar 2026 **Cell Cycle Conference 2026**, Online Conference.
Talk entitled *A genome language model for mapping DNA replication origins*.

Mar 2026 **Pathology Annual Symposium 2026**, Department of Pathology, University of Cambridge.
Plenary talk entitled *A genome language model for mapping DNA replication origins*.
★ *Runner-up, Best Talk Award.*

Dec 2025 **Research in Genetics Day 2025**, Department of Genetics, University of Cambridge, Cambridge.
Poster entitled *A computational framework for mapping DNA replication kinetics*.
★ *Awarded Best Poster Prize.*

Aug 2025 **Birmingham Medical School Research Seminar**, University of Birmingham, Birmingham.

Mar 2025 **CONNECTS-UK STEM Research Pitch Competition**, Jury Member, Corpus Christi College, Cambridge. Assessed research pitches from STEM researchers based in the United Kingdom and EU.

Nov 2024 **Research in Genetics Day 2024**, Department of Genetics, University of Cambridge, Cambridge.
Poster entitled *DNA replication timing reveals features of transcription and fragility*.

Oct 2024 **LUSO 2024**, PARSUK, London. Research pitch presenting current work to the Portuguese Association of Students and Researchers in the United Kingdom.

Sep 2024 **UK DNA Replication Meeting 2024**, UK DNA Replication Community, Churchill College, University of Cambridge. Poster entitled *A Whole-Genome Mathematical Model of DNA Replication*.

May 2024 **Mathematics and Applications Colloquium**, Applied Mathematics and Computational Science, CEMSE, KAUST.

Mar 2024 **Pathology Annual Symposium 2024**, Department of Pathology, University of Cambridge, Li Ka Shing Centre, Cancer Research UK. Plenary talk.

Mar 2024 **MCR–SCR Research Talks**, Queens' College, University of Cambridge, Cambridge. Research presentation to the college community.

Mar 2024 **Genetics Internal Seminar Series**, Department of Genetics, University of Cambridge, Cambridge. Internal research seminar.

Jan 2024 **Mathematical Biology Seminar**, Department of Mathematics, University College London, London. Research seminar.

Nov 2023 **Research in Genetics Day 2023**, Department of Genetics, University of Cambridge, Cambridge. Poster entitled *A Whole-Genome Mathematical Model of DNA Replication*.

Jul 2023 **Oxford Machine Learning Summer School 2023**, AI for Global Goals, Mathematical Institute, University of Oxford.
Advanced training in statistical and probabilistic machine learning, representation learning, reinforcement learning, causal inference, computer vision, and natural language processing.

Oct 2022 **Cellular and Molecular Pathology Seminar Series**, University of Cambridge, Cambridge.
Talk on Notch–Delta long-range signalling dynamics mediated by cellular protrusions.

Sep 2022 **ECMTB 2022**, European Conference on Mathematical and Theoretical Biology, Heidelberg University. Contributed talk in the mini-symposium *Bistable genetic switches across time, space, and disciplines*, entitled *The dynamics of long-range signalling via the Notch-Delta pathway*.

Apr 2022 **Henrik Jönsson Group Meeting**, Sainsbury Laboratory, University of Cambridge, Cambridge.
Invited talk entitled *Long-range signalling and patterns in the fly wing*.

Nov 2021 **Institute for the Physics of Living Systems**, University College London, London.
Talk and discussion entitled *How stretch affects patterning in the fly wing*, comparing mathematical models of signalling pathways and stretch-induced deformation of the wing disc.

Nov 2021 **Departmental Computational Biology Group Meeting**, University College London, London.
Research presentation to a multidisciplinary computational biology group, under the supervision of Philip Pearce.

Oct 2020 & Oct 2021 **Poster Competition Panel Member**, Pure and Applied Math Undergraduate Projects, UCL.
Supervision of a poster competition on differential geometry and dynamical systems applied in pandemic modelling.

April 2020 **Optimisation Article**, Chalkdust Magazine.
Published article on the inter-university mathematical journal *Chalkdust*, on a brief optimisation problem.

Mar 2014 – Jul 2014 **Internship**, Visiting Student Research Program, KAUST, Saudi Arabia.
Undergraduate internship and research project with a focus on Mean-Field Games, PDE and Functional Analysis. Supervised by Prof. Diogo A. Gomes.

Tests & Certifications

Sep 2018 – May 2019 **London Mathematical Society Courses**, Attendance and completion of four mathematical courses as part of the London Taught Course Centre (LTCC), via the London Mathematical Society. These included: Dynamical Systems, Stochastic Processes, Models and Measure Theory.

Dec 2018 **Teaching Training**, Completion of the *Teaching Undergraduates Mathematics and Statistics* (TUMIPS) training program at the Institute of Education, University College London.

Oct 2017 **TOEFL**, Score: 108/120.

Skills & Abilities

Programming Python, PyTorch, MatLab, R, Wolfram Mathematica, C, C#, C++, Beacon Calculus, DNAscent, Batch Programming, Nextflow.

Software Writing: LaTeX, MS Office. Teaching: Crowdmark, Blackboard Learn, STACK, WISEflow, Inspera.

Languages Portuguese (Native Speaker), English (fluent), comfortable with Spanish and French.

Extracurricular Activities

2025–Present **Ambassador, Portuguese Association of Students and Researchers in the UK (PARSUK)**.
PARSUK Ambassador, helping strengthen the network and activities of Portuguese researchers and students in the UK, fostering connections and supporting collaborative initiatives.

2023–2025 **Presidency, Vice-presidency, and Affiliation, Cambridge University Portuguese Society**.
Former president and vice-president of the Cambridge University Portuguese Society (CUPor), leading a dynamic cultural community, coordinating a diverse range of events, managing an executive committee, organizing sociocultural events, and advocating for Portuguese students at the university and beyond. Now active as a regular member.

2024–Present **Affiliate Researcher, Cambridge-Africa Programme**.
Member of the Cambridge-Africa Researchers' Database, supporting equitable research collaborations and capacity building with African institutions, and engaged in mentoring opportunities with African students and researchers.

2018–2022 **Student Union, University College London**.
Member of the Classical Guitar, Chamber Music and Hiking Societies at UCL.

2005–Present **Classical Guitar**.
Experience in playing various styles including Classical, Jazz, and Bossa Nova. Contributed to group musical arrangements, teaching via the Art's Office (KAUST), and participated in solo and ensemble performances.

2005–2012 **Almada's Music Academy, Portugal**.
Completed undergraduate studies in classical guitar (8th degree). Participation in several guitar contests and talent shows, winning first and second prizes, as well as honorable mentions.

2012–2016 **Volunteering**.
Participation in several volunteering activities, such as elderly home visits, beach cleaning and organization of homeless dinners. Leadership in a non-governmental volunteering organization in workshops in Portugal and Germany.

2005–Present **Sports**.
Amateur level sports, including football, tennis, swimming and skiing. 2 years of football at federate level.